



Increased Intracranial Pressure (ICp)

Prepared by /

Dr /Ghada elsayed

Outlines

❖ Introduction

❖ Definition

❖ Pathophysiology

❖ Causes of Increased Intracranial Pressure

❖ Assessment

❖ Management

Pre test



Mention causes of increase ICP



Mention symptoms associated with increase ICP

Increased Intracranial Pressure (ICP)

Many neurological problems are associated with increased ICP, such as brain injury and stroke.

Sustained increases in ICP compound the extent of brain injury and can be life-threatening.

Therefore, it is important to assess and maintain ICP within normal limits.

Normal ICP ranges from 0 to 15 mm Hg.



Definition

Increased ICP is a pressure of 20 mm Hg or greater persisting for 5 minutes or longer.



Pathophysiology

- The rigid cranial vault contains three types of non-compressible contents:
 - **brain tissue, blood, and CSF.**
 - The pressure exerted by the combined volumes of these three components is ICP.
 - If the volume of any one of these components increases, the volume of one or both of the other compartments must decrease proportionally or an increase in ICP occurs.
-

- 
- This compensation is termed compliance with adequate compliance; an increase in intracranial volume is compensated by displacement of CSF into the spinal subarachnoid space, displacement of blood into the venous sinuses, or both.
 - The ICP remains normal despite increases in volume.
-

As compensatory mechanisms are exhausted, a small increase in volume leads to a large increase in ICP.



As compliance decreases, ICP increases and Cerebral Blood Flow (CBF) decreases.



When CBF decreases, the brain becomes hypoxic, carbon dioxide levels increase, and acidosis occurs.



In response to these changes, the cerebral blood vessels dilate to increase CBF.

- This compensatory response further increases intracranial volume, creating a vicious cycle that can be life-threatening.
- Sustained increases in ICP can lead to herniation, which is caused by shifting of brain tissue from an area of high
- pressure to one of lower pressure.
- Along with Mean Arterial Pressure (MAP), the ICP determines Cerebral Perfusion Pressure (CPP), which is the pressure required to perfuse the brain.
- Cerebral Perfusion Pressure is calculated as the difference between MAP and ICP ($CPP = MAP - ICP$).

Normal CPP in an adult is between 60 and 100 mm Hg; it must be maintained at 70 mm Hg .

Any factor that decreases MAP and/or increases ICP decreases the CPP.

Ischemia or infarction can occur if the CPP is inadequate.

Measures to promote adequate CPP include lowering of increased ICP or, if this is not possible, increasing MAP to balance the effects of ICP on CPP.

Causes of Increased Intracranial Pressure

cerebral edema It is an increase in water content of brain tissue, if uncorrected; it may lead to increase ICP.



increased Cerebral Blood Volume (CBV), and increased CSF



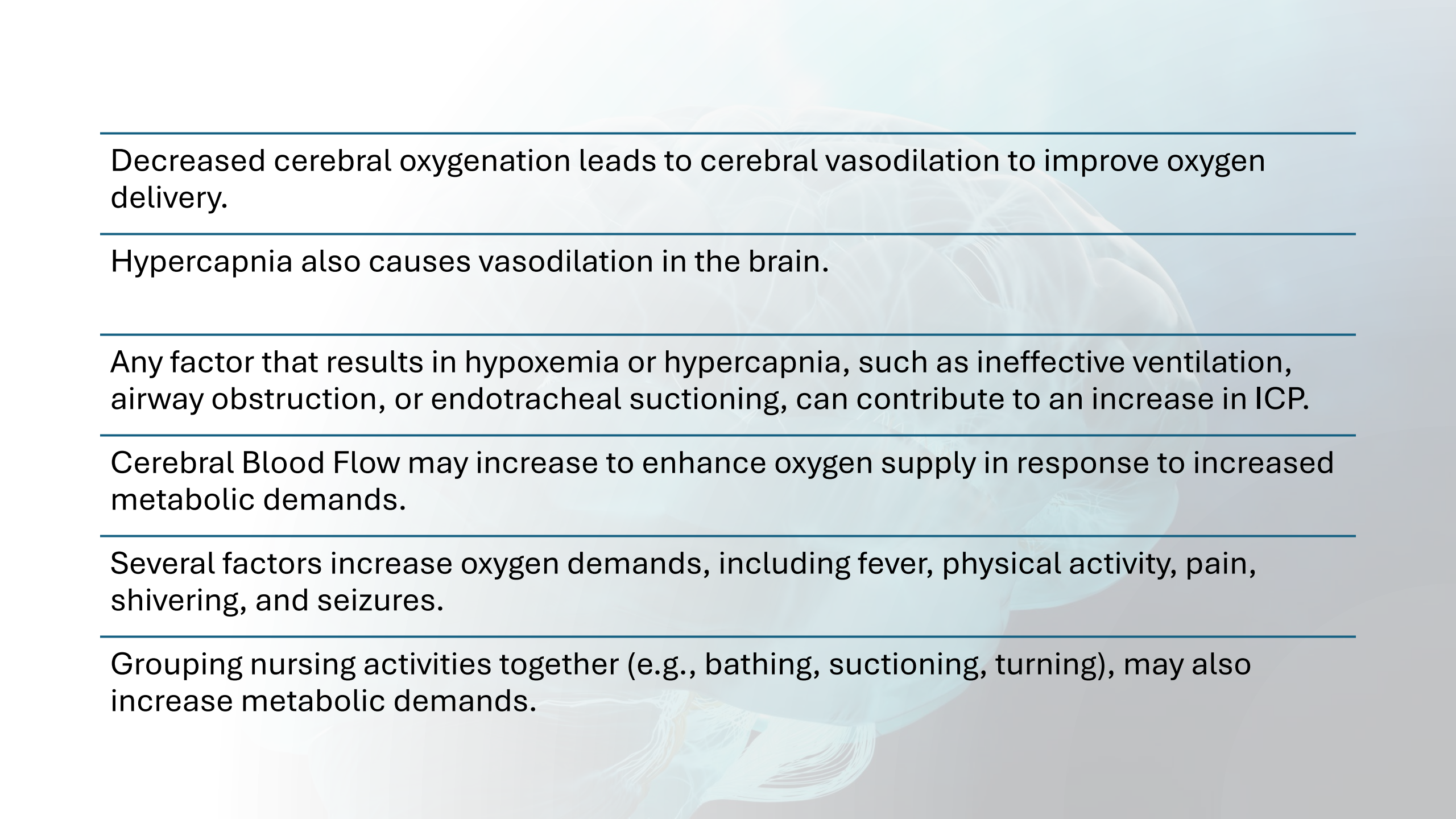
Head injury



Hypertension



Intracranial hemorrhage



Decreased cerebral oxygenation leads to cerebral vasodilation to improve oxygen delivery.

Hypercapnia also causes vasodilation in the brain.

Any factor that results in hypoxemia or hypercapnia, such as ineffective ventilation, airway obstruction, or endotracheal suctioning, can contribute to an increase in ICP.

Cerebral Blood Flow may increase to enhance oxygen supply in response to increased metabolic demands.

Several factors increase oxygen demands, including fever, physical activity, pain, shivering, and seizures.

Grouping nursing activities together (e.g., bathing, suctioning, turning), may also increase metabolic demands.

Increased Cerebrospinal Fluid: Hydrocephalus, an increase in CSF, can increase ICP.

Hydrocephalus may occur in any condition in which CSF flow or absorption is blocked, such as subarachnoid hemorrhage (SAH) or infection (meningitis, encephalitis), or when excess CSF is produced.

Assessment of Increased Intracranial Pressure Perform a careful neurologic examination

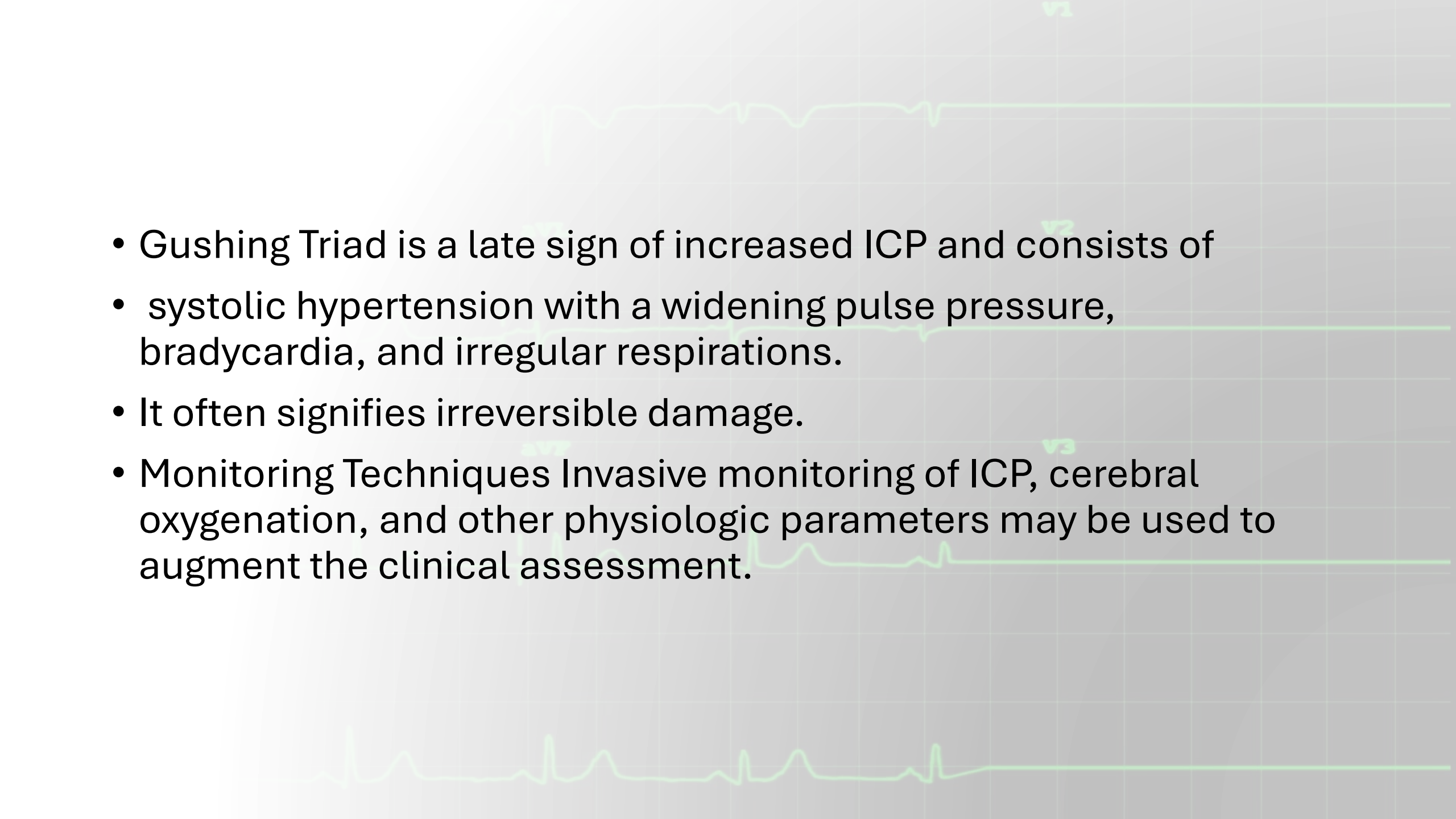
- with emphasis on Level Of Consciousness (LOC) and motor and cranial nerve function.
- Vital signs are part of the routine assessment.
- Hypertension is common early sign with increased ICP and represents a compensatory mechanism to augment Cerebral Perfusion Pressure (CPP).
- Increased ICP is manifested by deterioration in all aspects of neurological functioning.
- Level of consciousness decreases as ICP rises.

Initially, the patient may present with evidence of restlessness and confusion.

This then decompensate into lower levels of consciousness, ranging from lethargy to coma.

Pupillary reactions begin to diminish, and change in pupillary size.

Motor function also declines, and the patient begins to show abnormal motor activity (abnormal flexion or extension).

- 
- Gushing Triad is a late sign of increased ICP and consists of
 - systolic hypertension with a widening pulse pressure, bradycardia, and irregular respirations.
 - It often signifies irreversible damage.
 - Monitoring Techniques Invasive monitoring of ICP, cerebral oxygenation, and other physiologic parameters may be used to augment the clinical assessment.

Intracranial Pressure Monitoring:

- ICP monitoring is used to correlate objective data with the clinical picture and to determine cerebral perfusion.
- Cerebral Oxygenation Monitoring: CBF and brain oxygen utilization may be monitored by jugular oxygen saturation (SjO₂).
- Other Physiologic Monitoring: Hemodynamic monitoring may be used to monitor fluid status and to assist in maintaining adequate cerebral perfusion.
- Using pulse Oximetry for continuous monitoring of arterial oxygen saturation Continuous bedside electroencephalographic (EEG) .



Diagnostic Testing



Arterial Blood Gases (ABGs), SpO₂, end tidal carbon dioxide.



Complete Blood Count (CBC), with an emphasis on hematocrit, hemoglobin, and platelets.



Coagulation profile, including prothrombin time, International Normalized Ratio (INR), and activated partial thromboplastin time (a PTT) because brain injury may induce a coagulopathy
Electrolytes: Blood Urea Nitrogen, Creatinine, liver function and serum osmolality.

Radiological studies and other diagnostic tests that may be performed on patient with ICP as:

Computed Tomography (CT);

Magnetic Resonance Imaging (MRI);

CBF monitoring

EEG.

Management of Increased Intracranial Pressure

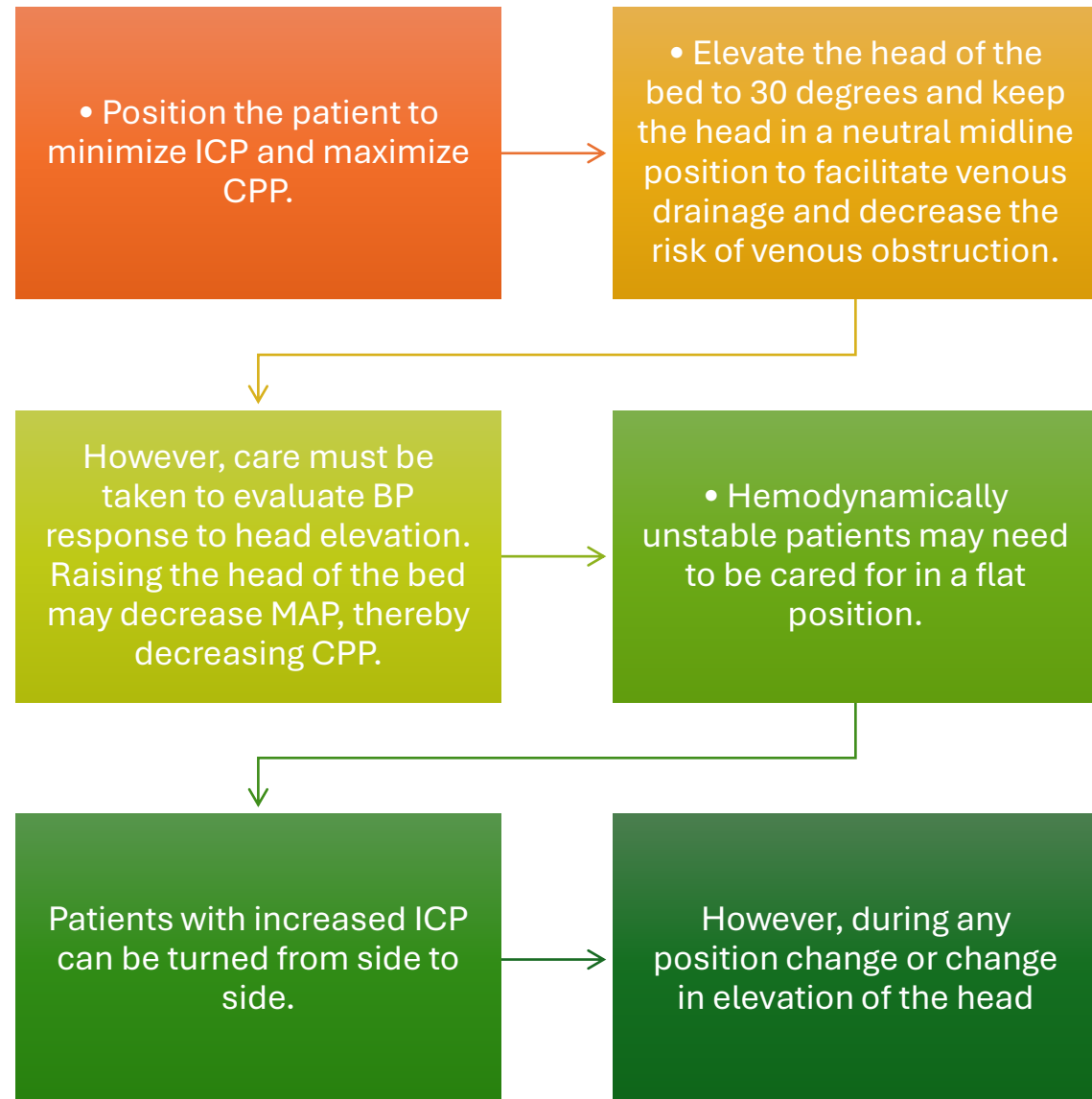
Medical and Nursing Interventions (Nonsurgical)

The goal of management is to maintain an ICP of less than 20 mm Hg while maintaining the CPP 60 to 70 mm Hg.

If the ICP is elevated, therapy is instituted to decrease ICP and then identify the cause of increased ICP.

Once the cause is discovered, management is centered on permanently decreasing the high ICP, maintaining CPP, maintaining the airway, providing ventilation and oxygenation, and decreasing the metabolic demands placed on the injured brain.

Nursing Management of Increased Intracranial Pressure



It is important to monitor and document the patient's individualized cerebral and hemodynamic response.

- If the CPP does not return to the baseline value within 5 minutes after the position change, reposition the patient to the position that maximized CPP.

- Suction the patient only when necessary because endotracheal suctioning is associated with hypoxemia.

Preoxygenate the patient with 100% oxygen before, between suction attempts and for 1 minute after the procedure.

Limit each suction pass to less than 10 seconds, with no more than two suction passes.

Maintain the head in a neutral position during the suctioning procedure.

-
- Spacing nursing care activities allows for rest between activities because several nursing activities are associated with increases in ICP as (turning, repositioning, and hygiene care).

Elevated ICP resulting from nursing care is usually temporary, and the ICP should return to the resting baseline value within a few minutes.



Assess the patient's physiologic response to visitors; family presence has been shown to decrease ICP.

However, caution family members to avoid excess stimulation of the patient or unpleasant conversations that may emotionally stimulate the patient.

Medical Management of Increased ICP

Medical management of increased ICP

Corticosteroids

Adequate oxygenation and ventilation

- Osmotic and loop diuretics
 - Fluid administration
 - Maintenance of BP
 - Reduction of metabolic demands.
-



-
- **Corticosteroids** are useful for reducing cerebral edema associated with brain tumors and meningitis.
 - Adequate Oxygenation and ventilation: The goal is to maintain a Pao2 above 80 mm Hg and to ensure that oxygen delivery to the brain exceeds oxygen consumption.
 - A Pao2 below 50 mm Hg can precipitate increased ICP.
 - For many patients with increased ICP, short-term management of the airway is accomplished by an end tracheal tube and mechanical ventilation.
 -

Positive end expiratory pressure may be added to facilitate oxygenation; however, it must be used with extreme caution because it may prevent venous outflow and further increase ICP.

A tracheostomy tube may be required for long-term ventilator management.

In addition, adequate hematocrit and hemoglobin levels are maintained to promote oxygenation.

Management of Carbon Dioxide:

Hyperventilation decreases PaCO_2 , which causes vasoconstriction of the cerebral arteries and a reduction of CBF.

Hyperventilation is used to decrease ICP for short periods when acute neurologic deterioration is occurring (i.e., herniation).

If the PaCO_2 level is lowered to less than 35 mm Hg for an extended period, Oxygen delivery at the cellular level should be evaluated with the use of a jugular bulb or cerebral tissue oxygen monitor.

Osmotic and loop diuretics: Osmotic and loop diuretics are administered to reduce cerebral brain volume by removing fluid from the brain's extracellular compartment.

Osmotic diuretics, such as Mannitol or hypertonic saline draw water from the brain to the plasma by creating an osmotic gradient, thereby decreasing ICP.

The effects of decreasing ICP and increasing CPP occur within minutes after the infusion.

Mannitol is given at a dose of 0.5-1 g/kg IV over 5-10 min, then 0.25- 2.0 g/kg IV bolus every 4-12 hr as needed depending on ICP. Side effects of osmotic diuretics include hypotension, electrolyte disturbances, and rebound increased ICP.

If Mannitol is used, the patient must have adequate intravascular volume to prevent hypotension and secondary brain injury.

Mannitol is contraindicated in patients with acute kidney injury because it is not metabolized and is excreted unchanged in the urine.

Loop diuretics (Furosemide, torsemide, bumetanide, ethacrynic acid) decrease ICP by removing sodium and water from injured brain cells. These agents also decrease CSF formation.

Optimal Fluid Administration: Fluid administration is provided to optimize MAP, maintain intravascular volume, and normalize CPP.

Normal saline solution (0.9%), an isotonic solution, is recommended for volume resuscitation.

Hypotonic solutions are avoided to prevent an increase in cerebral edema.

Strict measurement of intake and output while monitoring serum sodium, potassium, and osmolarity is required.

Colloids or blood products are administered if needed, to restore volume and maintain adequate hematocrit and hemoglobin levels. Hemodynamic monitoring may be used to optimize fluid administration.

Blood Pressure Management: BP must be carefully controlled in a patient with increased ICP.

Usually, the MAP is kept between 70 and 90 mm Hg. However, it is critical to monitor the ICP and MAP collectively to sustain an adequate CPP of at least 70 mm Hg.

-
- Hypotension decreases CBF, which leads to cerebral ischemia.

When hypotension occurs or ICP cannot be reduced, manipulating the systolic BP with vasopressor medications and fluids may be needed to achieve an adequate CPP.

- Hypertension (>160 mm Hg systolic) can worsen cerebral edema.

If necessary, systolic BP is lowered with antihypertensive medications such as labetalol.

This Beta Blocker decreases the sympathetic response and catecholamine release associated with neurologic injury.



Nicardipine is a calcium channel blocker medication that does not affect cerebral vasculature; it is very effective in providing faster and tighter control of BP than other antihypertensive medications.



Reducing Metabolic Demands: Several therapies may be required to reduce metabolic demands.



These include temperature control, sedation, seizure prophylaxis, neuromuscular blockade, and barbiturate therapy.



Temperature Control: Lowering body temperature to normal from a hyperthermic state or inducing hypothermia decreases cerebral metabolism, blood flow, and volume, thereby decreasing ICP.



Body temperature may be controlled noninvasively with the use of a cooling blanket or skin pads placed in direct contact with the skin (external cooling) or invasively by using a catheter placed in a large vein (intravascular cooling).



Target temperatures for induced hypothermia range from 34° to 35 C; these temperatures are maintained for 24 to 72 hours, followed by gradual rewarming.



Sedation: Patients are often sedated to lower increased ICP related to agitation, restlessness, or resistance to mechanical ventilation.

Benzodiazepines are given for sedation and do not affect CBF or ICP. Propofol is a sedative-hypnotic medication that reduces cerebral metabolism and ICP.

It is a short-acting medication with a rapid onset that is given by continuous infusion.

Morphine or fentanyl can be administered for analgesia and sedation in a low- dose continuous infusion or in small frequent intravenous (IV) boluses.

Neuromuscular Blockade and Barbiturate Therapy:
Neuromuscular blockade is considered for patients unresponsive to other treatments.

Barbiturates are given selectively to reduce ICP refractory to other treatments.

Pentobarbital is the most common medication used.

Patients receiving neuromuscular blockade or barbiturate therapy require hemodynamic monitoring, mechanical ventilation, and intensive nursing management.

Surgical Interventions: Surgical intervention may be required to remove a mass or lesion that is causing the increased ICP.

Post test

Mention causes of
ICP

Mention diagnostic
test for diagnosis
increase ICP



Any question



Thank you